Fiscal Year:	FY 2010	**Task Last Updated:**	FY 09/14/2010
PI Name:	Bloomberg, Jacob J. Ph.D.		
Project Title:	Development of Countermeasures to Enhance Sensorimotor Adaptation		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
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Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Cohen, Helen (Baylor College of Medicine) Reschke, Millard (NASA Johnson Space Center) Mulavara, Ajitkumar (USRA) Buccello-Stout, Regina (The University of Texas Medical Branch) Peters, Brian (Wyle Laboratories)		
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Task Description:

During adaptation to novel gravitational environments, sensorimotor disturbances have the potential to disrupt the ability of astronauts to perform required mission tasks. The overall goal of this project is to gain the information necessary to develop a comprehensive sensorimotor adaptability (SA) training program to facilitate rapid adaptation to novel gravitational environments. It has been previously shown that subjects trained to adapt to varied sensorimotor challenges can adapt faster to new sensory environments that they have never experienced before. This is a process known as adaptive generalization and it allows you to enhance the ability to "learn how to learn" to adapt to novel environments. By applying these motor-learning concepts for training astronauts we can enhance their ability to rapidly adapt their behavioral responses following a gravitational transition. To minimize cost and demands on crew time we have integrated SA training with existing exercise activities, namely treadmill walking. The SA training program we are developing entails manipulating the sensory conditions of treadmill exercise to systematically challenge multiple sensorimotor systems while conducting nominal exercise activities. To provide SA training we have mounted a treadmill on a six degree-of-freedom motion base to produce variation in the support surface along with variation in visual input during walking using both a projected virtual scene that produces variation in visual flow or through goggles that distort visuomotor information.

This project is in its second year. In our first study we aimed to determine whether sensorimotor adaptability (SA) skills learned during treadmill SA training could transfer to a novel discordant sensory environment and to quantify the extent to which any training effects would be retained. Twenty subjects (10 Training, 10 Control) completed three 30-minute training sessions consisting of various congruent and incongruent combinations of support surface and visual scene manipulations. Control group participants walked on the treadmill but did not receive any visual or support surface alterations. To determine the efficacy of training all subjects performed the Transfer/Retention Test upon completion of training. For this test subjects were exposed to novel visual flow and support surface movement, not previously experienced during training. The Transfer/Retention Test was performed 20 minutes, 1 week, 1, 3 and 6 months after the final training session. Stride frequency, auditory reaction time, and heart rate data were collected during this test as measures of postural stability, dual tasking ability, and anxiety, respectively. Subjects who received SA training showed less alteration in all three performance metrics compared to controls. We conclude that subjects who received SA training adapted faster than controls when presented with a novel discordant environment because they were able to apply adaptive skills that were learned during their earlier training sessions. Importantly, the training improved performance across a number of modalities including enhanced locomotor function, increased multi-tasking capability and reduced anxiety during adaptation to novel discordant sensory information. The Transfer Test was then re-administered to both groups 1 week, 1 month, 3 months, and 6 months later to examine retention of training. Trained subjects maintained their level of performance over six months.

In a second study we investigated whether performance in a novel discordant sensory environment can be predicted based on a subject's inherent visual dependency. To test individual visual dependency, subjects (n=10) were presented with incongruent visual flow during treadmill walking. Visual dependency was assessed by measuring the magnitude of torso translation during scene motion using video motion capture techniques. Following the visual dependency assessment subjects were exposed to a novel combination of support surface movement and visual flow. Subjects with greater visual dependency had increased stride frequency, reaction times and heart rates indicating that these subjects had decreased postural stability, increased anxiety, and decreased ability to dual task when negotiating novel sensory discordant conditions. These data indicate that visually dependency may be a "marker" for decreased ability to adapt to novel environments. Identifying preflight predictors of sensorimotor adaptability can be used to develop individualized training prescriptions that target the specific needs of each crewmember thus making the training process targeted and more efficient.

The goal of our third study (currently in progress) is to determine if SA training using treadmill walking transfers to other functional tasks including manual control. In this study subjects (n=45) are assigned to one of three groups. Subjects in the Treadmill Training group walk on a treadmill while wearing a series of goggles that distort visual input while viewing a virtual hallway scene thereby producing a sensorimotor adaptive challenge during treadmill exercise. Subjects in the two other groups serve as controls. Before experiencing the training subjects perform two operationally oriented manual control tests (Purdue Pegboard Test, Construction Activity Test). After completing the training, all subjects (regardless of group) perform both manual control tests while wearing goggles with unique visual distortions not previously experienced by the subjects to determine if SA training has transferred from the locomotor system to the manual control system. These data will help us determine if treadmill SA training can generalize beyond locomotion. If it does transfer to other functional tasks we will need only one type of training modality to enhance other functional tasks including those comprised of complex manual control requirements.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Sensorimotor adaptability training programs have earthbound application in rehabilitation of patients with balance disorders, and for fall prevention training among seniors. The most important factors underlying morbidity in the older adult population are injurious falls and the restriction of activity due to falls. Approximately 25-35% of community-dwelling persons older than 65 years fall at least once a year, and approximately 40-50% of fallers experience two or more falls. The outcome is that 40% of all nursing home admissions are due to injuries due to falls. Age-related sensory changes and deterioration in ability to compensate in older adults can contribute to increase incidence of falling.

We previously conducted a study to determine if balance training using variation in visual flow during treadmill exercise improves functional mobility in healthy older adults who were experiencing age-related postural instabilities (Buccello-Stout et al. 2008). The results showed that subjects who were exposed to varied visual flow during treadmill walking significantly improved their ability to negotiate an obstacle course after training compared to another elder group who only walked on a treadmill for the same amount of time. Importantly, the training benefit was retained when subjects were tested four weeks later. This study confirms that adaptability training developed for use by astronauts can also be used to improve balance and gait performance in elder subjects and points to the general applicability of this type of training in different clinical populations.

Task Progress:	1) Completed the data collection and analysis for the first two studies in this project. The first study determined the transfer efficacy along with the short (days) and long (months) term retention properties of our sensorimotor adaptability (SA) training program. The second study focused on visual dependency as a potential predictor of individual ability to adapt to novel discordant sensory stimuli. 2) Completed 8 Abstracts for scientific meetings and 2 papers have been submitted for publication. 3) Initiated a new study that is currently in progress to determine if SA training using treadmill walking transfers to other functional tasks including manual control.
Bibliography Type:	Description: (Last Updated: 11/14/2025)
Abstracts for Journals and Proceedings	Batson CD, Brady RA, Peters BT, Mulavara AP, Bloomberg JJ. "Gait training improves performance in healthy adults exposed to novel discordant conditions." 3rd International Congress on Gait and Mental Function, Washington, D.C., February 26-28, 2010. 3rd International Congress on Gait and Mental Function, Abstract 102, Abstract Book, February 2010. , Feb-2010
Abstracts for Journals and Proceedings	Bloomberg JJ, Mulavara AP, Peters BT, Brady R, Batson C, Cohen HS. "Sensorimotor adaptability training improves motor and dual-task performance." NASA Human Research Program Investigators' Workshop, Houston, Texas, February 3-5, 2010. NASA Human Research Program Investigators' Workshop, Abstract Book, February, 2010. , Feb-2010
Abstracts for Journals and Proceedings	Bloomberg J, Mulavara A, Peters B, Brady R, Batson C, Cohen H. "Training modalities to increase sensorimotor adaptability." 81st Annual Scientific Meeting of the Aerospace Medical Association, Phoenix, AZ, May 9-13, 2010. Aviation, Space, and Environmental Medicine 2010 Mar;81(3):215-6. , Mar-2010
Abstracts for Journals and Proceedings	Brady RA, Batson CD, Peters BT, Mulavara AP, Bloomberg JJ. "Sensory bias predicts postural stability, anxiety, and cognitive performance in healthy adults walking in novel discordant conditions." 3rd International Congress on Gait and Mental Function, Washington, D.C., February 26-28, 2010. 3rd International Congress on Gait and Mental Function, Abstract Book, February 2010. Abstract #101. , Feb-2010
Abstracts for Journals and Proceedings	Brady RA, Peters BT, Bloomberg JJ. "Walking on an oscillating treadmill: two paths to functional adaptation." The 34th Annual Meeting of the American Society of Biomechanics, Providence, Rhode Island, Aug.18-21, 2010. The 34th Annual Meeting of the American Society of Biomechanics. Abstract Book, August 2010. , Aug-2010
Abstracts for Journals and Proceedings	Buccello-Stout RR, Cromwell RL, Bloomberg JJ. "Characterization of the head stabilization response to a lateral perturbation during walking in older adults." Gerontological Society of America 62nd Annual Scientific Meeting, Atlanta, GA, November 18-22, 2009. Gerontological Society of America 62nd Annual Scientific Meeting, Abstract Book, November 2009. , Nov-2009
Abstracts for Journals and Proceedings	Buccello-Stout RR, Cromwell RL, Bloomberg JJ. "Functional mobility performance and balance confidence in older adults after sensorimotor adaptation training." Gerontological Society of America 63nd Annual Scientific Meeting, New Orleans, LA, November 19-23, 2010. Gerontological Society of America 63nd Annual Scientific Meeting, Abstract Book, Nov. 2010. , Nov-2010
Abstracts for Journals and Proceedings	Ruttley T, Holt C, Mulavara AP, Bloomberg JJ. "The role of adaptation in body load-regulating mechanisms during locomotion." The Society for Neurosciences 40th Annual Meeting, San Diego, CA, November 13-17, 2010. The Society for Neurosciences 40th Annual Meeting, Abstract Book, November 2010. , Nov-2010
Articles in Other Journals or Periodicals	Batson CD, Brady RA, Peters BT, Ploutz-Snyder RJ, Mulavara AP, Bloomberg JJ. "Gait training improves performance in healthy adults exposed to novel sensory discordant conditions." Experimental Brain Research. In press, 2010. , Sep-2010
Articles in Other Journals or Periodicals	Brady RA, Brian BT, Bloomberg JJ. "Walking on an oscillating treadmill: Strategies of stride-time adaptation." Gait and Posture. Submitted, 2010. , Aug-2010
Awards	Bloomberg J. "JSC Director's Innovation Group Achievement Award, May 2010." May-2010
Journal/Magazine covers	Medical Design Briefs. "Our Sensorimotor Adaptation Training System was featured and made the cover of Medical Design Briefs Magazine, 'Re-Learning Life: Rehabilitation Technologies are Key to Recovery.'" Medical Design Briefs, 2010, Jan. , Jan-2010
Significant Media Coverage	Salazar J. " 'Jacob Bloomberg helps astronauts get their balance back.' Participated in a podcast for the science website EarthSky describing our current NSBRI study." EarthSky. March 22, 2010. http://earthsky.org/space/jacob-bloomberg-helps-astronauts-get-their-balance-back , Mar-2010
Significant Media Coverage	Thomas B, Bloomberg JJ. "Completed a press release with NSBRI describing our Sensorimotor Adaptation Training System. The press release, 'Astronaut balancing act: Training to help explorers adapt to a return to gravity,' was picked by numerous news organizations." Science Daily, December 11, 2009: http://www.sciencedaily.com/releases/2009/12/091209134640.htm , Dec-2009